

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Machine Learning / Statistics:** Classification, Regression, Feature Engineering, Model Evaluation, Precision, Recall, Accuracy, Statistical Analysis, Predictive Analytics, Data Validation

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, Scikit-learn, LightGBM

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

• Designed and implemented end-to-end machine learning pipeline to ingest, process, and analyze sensor data from 100+ connected assets, combining scalable data ingestion, feature engineering, and model-ready transformations to enable predictive operational insights and downstream analytics

• Developed and optimized scalable data models and query execution plans across cloud and relational database environments, improving query performance by 38% while enabling efficient feature extraction for machine learning workflows

• Built embedding-based retrieval and summarization pipeline to enhance contextual understanding of user interactions, improving multi-turn response accuracy by 43% and supporting more reliable behavioral analysis

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Developed NLP and text analytics pipelines in Python to extract structured signals from large-scale unstructured data, applying text preprocessing, feature extraction, and model optimization to reduce processing latency by 41%

• Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets

• Designed production-grade inference **APIs** and deployment workflows to support reproducibility, monitoring, and consistent evaluation of machine learning outputs

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

• Built and deployed **Retrieval-Augmented Generation (RAG)** system using embeddings and **vector search**, combining data pipelines, feature engineering, and retrieval optimization to improve document retrieval efficiency by 64%

• Designed scalable **ETL** and feature pipelines using **FastAPI** and **PostgreSQL** to transform unstructured data into structured formats suitable for analytics and machine learning use cases

• Optimized indexing strategies and data processing workflows to enable low-latency querying and scalable deployment of AI-driven systems

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

• Analyzed large-scale logistics and operational datasets using Python and PySpark to identify trends, anomalies, and performance issues, applying exploratory analysis and root cause investigation to support supply chain decision-making

• Built Power BI dashboards to track key performance indicators and present operational insights to leadership, reducing reporting time by 40% and enabling data-driven business decisions

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Scientist

• Developed and maintained Power BI dashboards using **SQL** and **DAX**, transforming raw data into actionable insights for business decision-making across multiple domains.

• Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois

August 2025 - Present

Project Manager

• Applied AI-assisted analysis to 10000+ player feedback records and behavioral datasets, identifying patterns that accelerated feature prioritization by 35%

PROJECTS

Scalable ML System for Customer Risk Prediction

• Built LightGBM-based machine learning model to forecast outcomes from time-series transaction data, combining feature engineering, classification modeling, and business statistics techniques to identify and segment high-risk and low-risk patterns

• Performed feature engineering on behavioral patterns such as transaction frequency and spending trends, improving **model precision** by 20% while enhancing robustness and interpretability

• Evaluated model performance using precision, recall, and accuracy, and designed a decision framework to translate analytical predictions into actionable business recommendations for stakeholders